# Physics 9th – Chapter-9-Nature of Science and Physics (Short Response Questions)

**Q1. Define the following branches of physics: Biophysics, Astrophysics, Optics, Relativistic Mechanics, Nuclear Physics, Acoustics, Computational Physics.**

**Ans:**

- Biophysics: study of living systems using physics (e.g. blood flow, radiation in diagnosis). Astrophysics: study of the physical nature of stars and celestial bodies using laws of physics.
- Optics: study of the nature and properties of light and instruments that use it. Acoustics: production, transmission, reception and effects of sound.
- Relativistic mechanics: motion of bodies moving with speeds close to the speed of light. Nuclear physics: structure, properties and behaviour of atomic nuclei.
- Computational physics: use of numerical analysis and computers to solve complex physics problems.

**Q2. Define the terms theory and law. Also give two examples of each.**

**Ans:**

- Theory: a well-tested, well-supported explanation of how and why nature behaves in a certain way under specific conditions.
- Examples of theories: Kinetic Molecular Theory, Theory of General Relativity.
- Law: a statement that summarises an observed regularity in nature and gives the relationship between variables.
- Examples of laws: Newton's laws of motion, Newton's law of universal gravitation (also Boyle's law).

**Q3. Define hypothesis.**

**Ans:**

- A hypothesis is a tentative assumption or explanation made before research, which can be verified by further investigation (experiments).
- It is a testable prediction that may be proved true or false.
- Concept: a hypothesis verified again and again can become a theory or a law.

**Q4. What is the difference between classical mechanics and quantum mechanics?**

**Ans:**

- Classical mechanics studies the motion of macroscopic (large) objects under the action of forces, e.g. motion of a car or a planet (Newton's laws).
- Quantum mechanics gives the mathematical description of the motion and behaviour of microscopic particles like electrons and atoms.
- Classical mechanics fails at the atomic scale, where quantum mechanics is needed.

**Q5. What determines the validity of a theory?**

**Ans:**

- A theory is valid if its predictions agree with the results of repeated experiments and observations.

- It must be testable and falsifiable, i.e. it can be checked and could be disproved by evidence.
- If new reliable evidence contradicts it, the theory is modified or replaced (e.g. Newton's laws improved by Einstein's relativity).

**Q6. Which part of the study is not handled by the classical study of physics?**

**Ans:**

- Classical physics cannot explain the behaviour of microscopic particles (atoms, electrons, nuclei); this is handled by quantum mechanics.
- It also fails for bodies moving with speeds close to the speed of light; this is handled by relativistic mechanics (Einstein's theory of relativity).
- For particles that are both very small and very fast, quantum field theory is used.